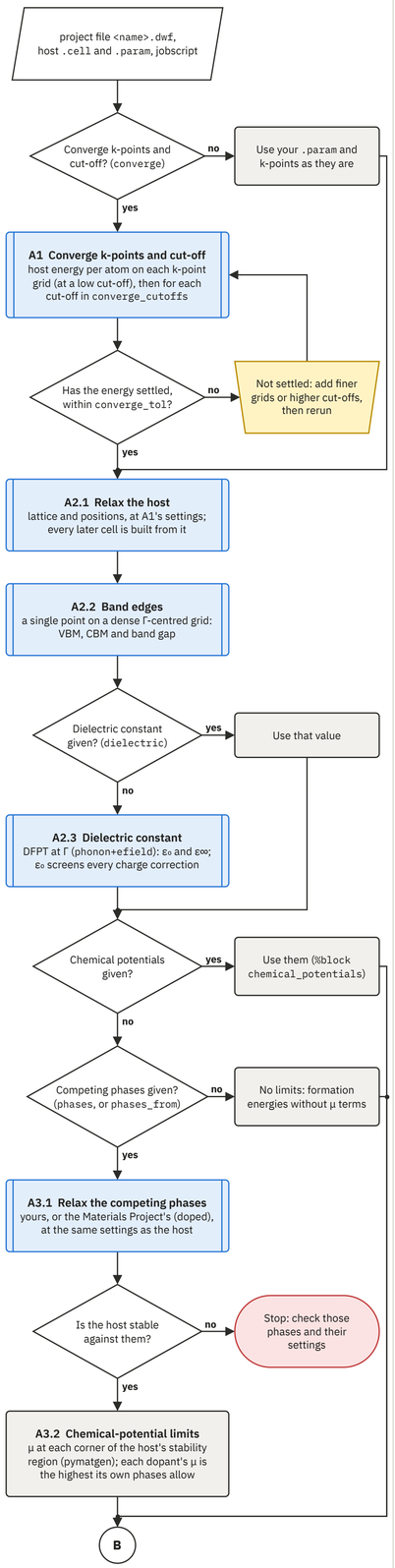
Defect workflow

Read each column from top to bottom, A to G: F and G share the last one. Diamonds are decisions, and every branch out of one is labelled; "given" means set in the project file. Dashed steps are planned, not scripted yet. One command runs every scripted step in order: `python3 dwf.py <name>.dwf`. Each phase writes summary.txt (for you to read) and results.json (for the later phases) in its own folder.

start / end · done by dwf.py itself · CASTEP jobs (run through supercell_convergence.py) · MACE (B2, C1: a foundation model; D3–D5: one trained per defect) · ShakeNBreak · needs your attention · decision · input or data · planned, not scripted yet · (B) continues in the next column

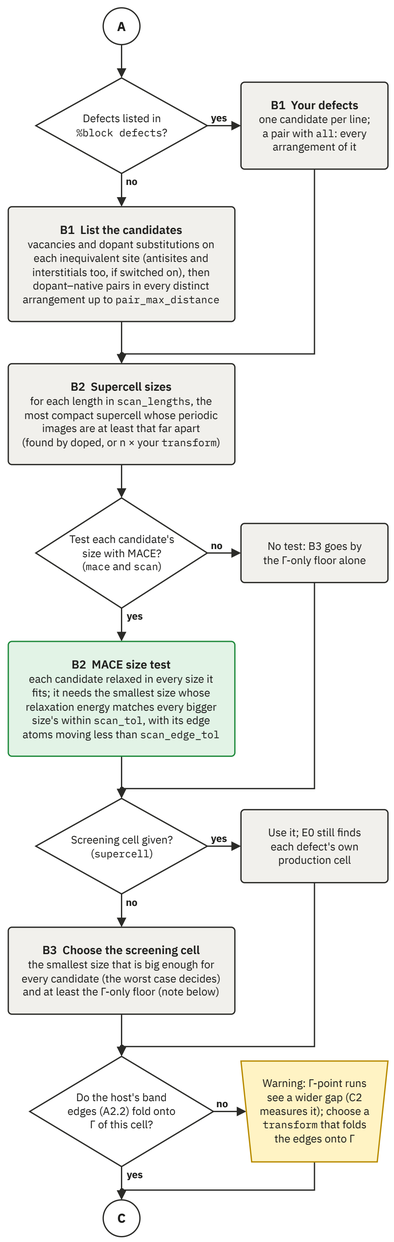
A Host settings and reference data

A_host/ · once per host



B Candidate defects and the screening cell

B_candidates/ · once per host



The Γ-only floor (B3)

C screens at the Γ point only, which gives a defect's levels as an isolated defect has them only when its periodic images barely overlap. A level E_b inside the gap decays as exp(−r/ξ), where (two-band model) $1/\xi^2 = (2m^*/\hbar^2)\, E_b\,(1 - E_b/E_g)$. The screening cell keeps its images at least 4ξ apart, with

- E_b = deep_level_margin: the shallowest level that counts as deep
- m*: the lighter of the hole and electron masses at A2.2's band edges
- E_g: A2.2's band gap

Set screen_min_image to give the floor yourself (0: no floor).

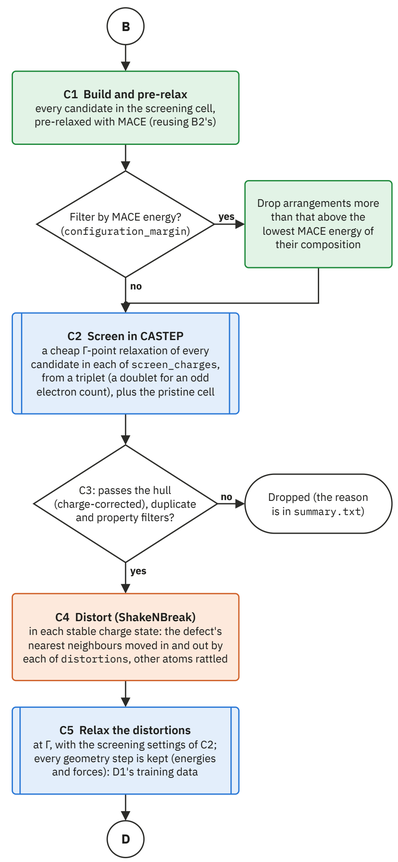
Cell sizes are image distances

A size is the shortest distance between a defect and its periodic images (Å), not an atom count. The sizes come from scan_lengths:

- B2: MACE tests every candidate in every size
- B3: the screening cell fits the worst case and the Γ-only floor
- E0: steps up the sizes to each defect's own production cell

C Screening

C_screen/ · every candidate, then the survivors



Until D1–D5 are scripted

dwf.py goes from C5 straight on to D6 (the solid arrow): D6 ranks C5's CASTEP relaxations. What it already keeps for the model:

- every geometry step of C5 (energies and forces), in each .castep
- E2: E0's and E1's steps as training data, charged ones with their total charge (charged.xyz)
- mace_model takes your own model file (B2 and C1, charge-blind)

The model in D1–D5: Wang et al.

Wang, Mosquera-Lois and Walsh, PRX Energy 2026 (arXiv:2603.05238): Sb_2Se_3, one 60-atom cell, VASP (PAW, 350 eV, DFT-D3).

- $V_{Se}(1)$ in q = −2 … +2: one charge-embedded MACE, trained on HSE06 Γ-point relaxations of ShakeNBreak distortions (2,287 structures)
- neutral $V_{Se}(1)$: a multi-fidelity MACE from 452 PBE steps and 34 HSE06 single points
- MACE 0.3.15 from scratch: 2 layers, 128 channels, 5.5 Å cut-off, ZBL
- test errors 0.5 meV/atom and 20 meV/Å; transition levels within 0.015 eV of HSE06; the search about 1,000 × cheaper than HSE06
- one model per defect, site and cell size; negative charge states were the hardest; the charge embedding and the two heads (D3) were never tested together: they trained them as two models

Data and trained models: Zenodo, record 22756145 (CC BY 4.0).

Foundation models on defects (their test)

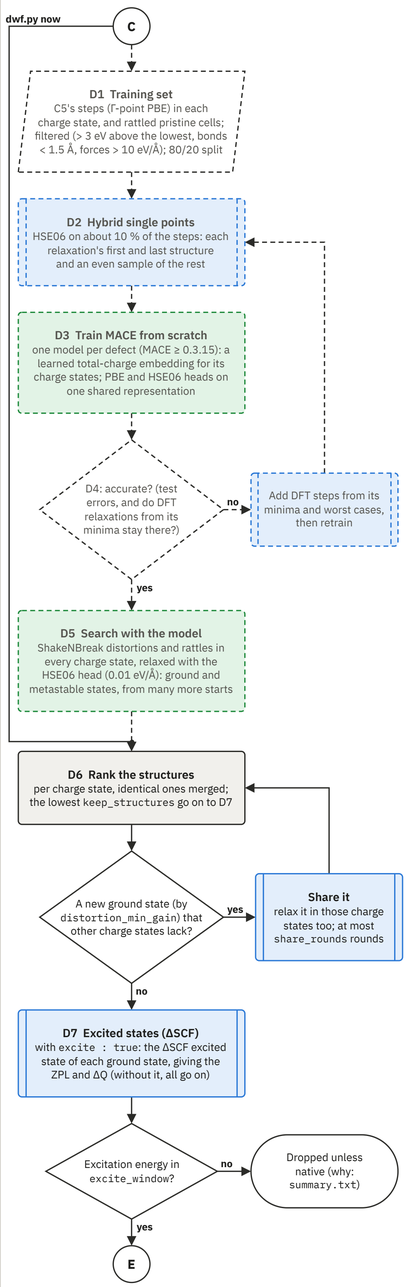
On $V_{Sb}(1)$ at the PBE level, MACE-MH-1, MACE-MPA-0, GRACE-2L-OMAT and MatterSim put the minima 0.2–0.4 Å from DFT's (0.1 Å counts as a match), on flatter energy surfaces, and DFT relaxations from their minima missed the DFT ground states. The charge-aware MACE-POLAR-1-L and QET-PES-MatQ failed too. So B2 and C1 keep a foundation MACE for cell sizes and pre-relaxation only, and each defect gets a model trained on its own DFT data (D3).

What the filters in C and D keep

- C1 configuration_margin: arrangements within that of the lowest MACE energy of their composition (MACE ignores the charge)
- C3 hull: an arrangement stays if, at some Fermi level in the gap, its lowest formation energy is within hull_margin of its group's
- C3 duplicates: one that relaxed into a lower one's structure goes
- C3 properties (optional), met by a charge state stable somewhere in the gap: require_gap_states, require_spin, require_bound
- D6: the keep_structures lowest structures of each charge state

D Ground and excited states

D_states/ · the survivors; D1–D5 (dashed) are not scripted yet



D7: excited states (`excite : true`)

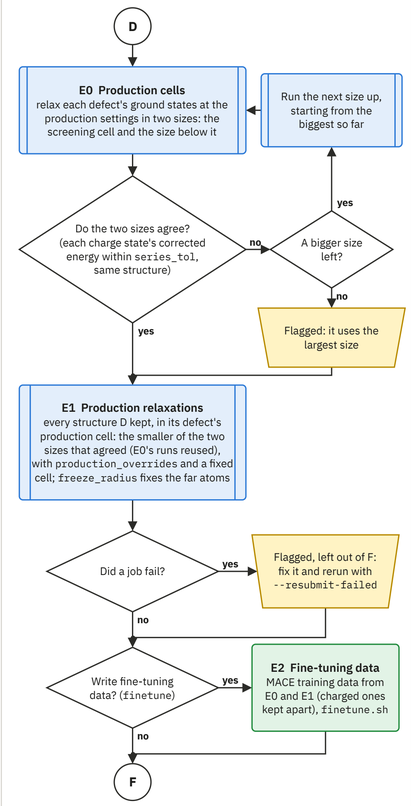
- ground state: a single point at the screening settings (metals_method : dm), keeping its .check
- excitation: the smallest step, at least excite_min, from an occupied to an empty gap level in one spin channel
- excited state: CASTEP's ΔSCF, started from that .check; relaxed (excite_relax) it gives the ZPL and ΔQ, else the vertical energy
- verdict: inside excite_window (pass), outside, no excitation, or unknown (its runs failed: kept, with a warning)
- native defects always go on, and skip C3's property filters

Test D7 on one defect first

CASTEP's ΔSCF must work on your build: test D7 first on one defect with a known ZPL (in CaO, Sb_{Ca}–V_O at q = −1: ADAQ gives about 0.53 eV). If its excited state won't relax, set `excite_relax : false`.

E Production cells and relaxations

E_production/ · each surviving defect, in every charge state



How the CASTEP jobs run

Every CASTEP step hands its jobs to `supercell_convergence.py`:

- one short dry run per distinct input sizes the jobs: cores or nodes, memory, partition and walltime; where nodes are shared and charged by the core (Cirrus), each job asks for just the cores it needs
- no more than max_jobs are queued at once
- a job that runs out of time carries on from its .check (up to max_restarts times; screening jobs don't restart)
- with tolerate_failures (screening, E0, E1), a failed job is flagged and the others carry on
- a finished job whose inputs have since changed stops the run, so results from two different settings never mix
- touch <workdir>/STOP stops dwf.py; queued jobs keep running

Stopping and rerunning

<workdir>/state.json records each finished step, with a fingerprint of its settings, input files and earlier results.

- stopped (STOP, Ctrl-C, --no-wait, a crash)? Run the same command again: finished steps are skipped, and jobs already submitted or finished are picked up
- changed a setting? The steps that depend on it run again; jobs whose inputs are unchanged are reused
- --redo STEP: start that step again (its old files are kept)
- --status: what has finished; --until B: stop after phase B
- --prepare-only: write the job inputs, but submit nothing

Charge corrections: where they're applied

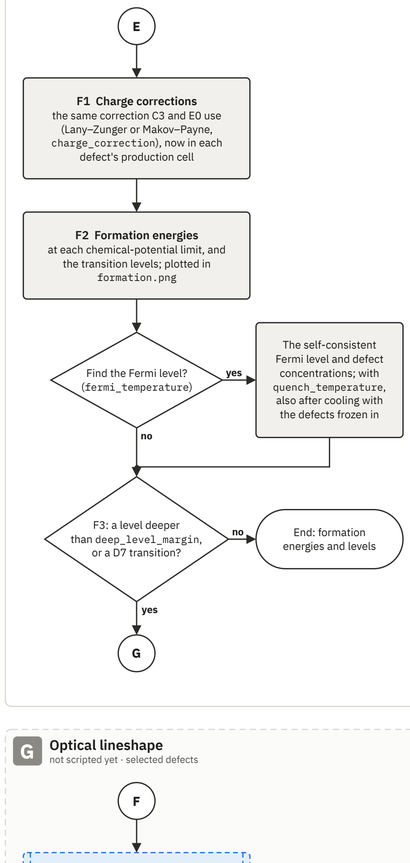
Lany–Zunger (default) or Makov–Payne: a point charge q screened by ε₀ (A2.3), in each cell's own shape (charge_correction).

- C3: the hull, stable charge states and binding (screening cell)
- E0: each size's $E_D - E_{host} + E_{corr}$, before sizes are compared
- F1–F2: each defect's production cell

Not included: potential alignment, an anisotropic ε (averaged).

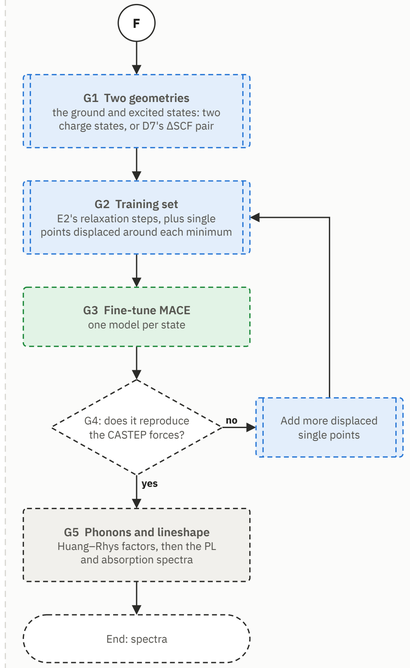
F Thermodynamics

F_thermo/ · every defect that reached E



G Optical lineshape

not scripted yet · selected defects



Phase G is not scripted yet

F3 lists the defects worth a lineshape (for_G in F_thermo/results.json), and E2 saves the E0 and E1 relaxation steps as training data for G2. Planned tools: MACE for fine-tuning and phonons, and lineshape_tools (Turiansky et al.) for the spectra.